MIL-R-43448A
13 December 1968
SUPERSEDING
MIL-R-43448
8 August 1966

MILITARY SPECIFICATION

REELS, STRAPPING COIL AND TRUCKS, HAND, STRAPPING COIL

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers strapping coil reels, and hand trucks for the dispensing of flat steel, round steel, flattened wire and nonmetallic strapping.

1.2 Classification.- Strapping coil reels and hand trucks covered by this specification shall be of the following types and classes:

Type I	-	Reels, strapping coil.
Class A	-	For dispensing oscillated wound flat steel strapping coils.
Class B	-	For dispensing nonmetallic strapping coils
Type II	-	Trucks, hand, strapping coil.
Class A	-	For dispensing oscillated wound flat steel strapping coils.
Class B	-	For dispensing nonmetallic strapping coils.
Class C	-	For dispensing round steel or flattened wire strapping coils.
Class D	-	For dispensing ribbon wound flat steel strapping coils.
Class E	-	For dispensing ribbon wound flat steel strapping coils, roller mounted.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:27Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

QQ-S-781	-	Steel, Strapping, Flat.
QQ-S-790	-	Steel, Strapping, Round (Bare and Zinc-Coated).
PPP-B-576	-	Boxes, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-591	-	Box, Fiberboard, Wood-Cleated.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Box, Fiberboard.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-S-760	-	Strapping, Nonmetallic, (and Connectors).

MILITARY

MIL-L-10547	-	Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-147	-	Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:27Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

* 3.1 First article.- Unless otherwise specified (see 6.2), before production is commenced, a preproduction sample of the strapping coil reel or hand truck covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

* 3.2 Standard product.- The strapping coil reels and hand trucks shall be the current commercial product of the manufacturer except for any changes necessary to comply with the requirements of this specification.

* 3.3 Design and construction.- Strapping coil reels and hand trucks shall be constructed of steel except as specified (see 3.3.2 and 3.3.3). The strapping coil reel and hand truck shall consist of a frame, means to facilitate loading, retain coils, dispense strapping as required, prevent strapping from being unraveled or unwound, provide mobility as applicable, and shall be capable of holding coils conforming to QQ-S-781, QQ-S-790 and PPP-S-760, as applicable.

3.3.1 Type I.- Type I strapping coil reels shall dispense a coil vertically mounted on the frame having a stable floor stand base.

3.3.2 Type II.- Type II strapping coil hand trucks shall have two wheels not less than 3 inches diameter, a tool and accessory tray attached to the frame and one or two handles to facilitate mobility. A pivit wheel may be provided at the option of the supplier. Unless otherwise specified (see 6.2), the wheels shall have rubber tires. Hand trucks except class E shall dispense strapping coils from a reel vertically mounted on the frame. The class E hand truck shall dispense ribbon wound flat steel strapping by rotation of the coil on rollers mounted at the bottom of the frame. The frame shall have racks capable of holding and dispensing multiple coils of varied sized strapping separately or at the same time.

3.3.3 Type I and II, Class B.- Type I and II, class B, strapping coil reel, and hand truck, shall have reel dispensers with flanges made of steel or plastic at the option of the manufacturer.

3.4 Threads.- All screw threads shall conform to Handbook H28.

* 3.5 Performance.-

* 3.5.1 Stability.- The type I reel loaded with a full coil of strapping shall withstand the drop test with no visible evidence of deformation, when tested as specified in 4.4.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:27Z
Check the source to verify that this is the current version before use.

3.5.2 Maneuverability.- The type II hand truck loaded with full coils of strapping shall be maneuverable with no binding or visible evidence of permanent deformation, when tested as specified in 4.4.2.

* 3.5.3 Dispensing.- All strapping coil reels and hand trucks shall be capable of dispensing strapping freely with no drag except for that supplied by the restraining means to control tension and overrunning of the coil, when tested as specified in 4.4.3.

3.6 Finish.- The strapping coil reel and hand truck shall be finished in accordance with the supplier's standard commercial practice.

3.7 Marking.-

3.7.1 Marking for identification.- The strapping coil reel and hand truck shall be marked in accordance with the manufacturer's standard commercial practice.

3.7.2 Loading instruction marking.- A loading instruction label shall be affixed to the reel and hand truck in a location that is readily seen. This label shall show the proper installation of the strapping coil.

* 3.8 Instruction sheet.- The supplier shall furnish with each strapping coil reel and hand truck, an instruction sheet. Contents of the instruction sheet and additional distribution shall be as specified in the contract or order (see 6.2).

* 3.9 Workmanship.- The strapping coil dispenser shall be clean and in alignment. All welds shall be continuous, adherent, and free from burn holes and flux deposits. The strapping coil dispenser shall be free of burrs and slivers. Painted surfaces shall be dry, completely covered and smooth, not crazed, and free from runs, sags, or unpainted areas. Rivets shall be tight and rivet heads fully formed and not cracked.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First article inspection.- When a preproduction sample is required, it shall be examined for defects in table I, and tested as specified in 4.4.1 or 4.4.2, as applicable and 4.4.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:27Z
Check the source to verify that this is the current version before use.

4.3 Sampling for inspection.- Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase doucment.

* 4.3.2 Inspection of the end item.- The lot shall be all strapping coil reels or hand trucks of one type and class offered for inspection at one time. The sample unit shall be one complete strapping coil reel or hand truck.

4.3.2.1 Visual examination.- The completely assembled strapping coil reel or hand truck shall be examined for defects in table I. The inspection level shall be level I with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

* TABLE I.- Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Painted surfaces crazed, not dry or smooth, not free from runs, sags, or unpainted areas.		X
Construction and workmanship (general) (applicable to all components and assemblies)	Part missing or not specified type or size.	X	
	Part fractured, split, bowed, malformed, or impaired.	X	
	Part missing, loose, or not in proper alignment.	X	
	Sharp burrs and slivers.	X	
Welding	Missing, not continuous and adherent, burn holes.	X	
	Flux deposits not removed.		X
Rivets	Loose.		X
	Heads not fully formed or cracked.	X	
Threaded fasteners	Missing, broken, stripped, fractured, or loose.		X

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:27Z
Check the source to verify that this is the current version before use.

TABLE I.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Identification marking	Missing, incomplete, not legible, or not as specified.		X
Instruction sheet	Missing, incomplete, not legible, or not as specified.		X

4.3.2.2 Dimension examination.- Examination shall be made of the end item to determine compliance with dimensional requirements. Any nonconforming dimensions shall be classified as a defect. The inspection level shall be level S-2 with an AQL of 6.5, expressed in terms of defects per hundred units.

4.3.3 Testing of the end item.- The completely finished strapping coil reel or hand truck shall be tested in accordance with 4.4.1 and 4.4.2, as applicable, and 4.4.3. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

4.3.4 Examination of preparation for delivery.- An examination shall be made to determine that packaging, packing, and marking as required by section 5 of this specification are complied with. Defects shall be scored as specified in table II. The lot shall be the number of containers offered for delivery at one time. The unit of product for sampling shall be one shipping container fully prepared for delivery, except it need not be sealed. The inspection level shall be S-2 and the AQL shall be 4.0, expressed in terms of defects per hundred units.

TABLE II.- Examination of preparation for delivery

Examine	Defect
Marking (exterior and interior)	Missing; incorrect; illegible; of improper size, location, sequence of method of application.
Materials	Component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of case liners or container flaps, loose strapping; inadequate stapling, bulging or distortion of containers; blocking, and cushioning inadequate.
Content	Content more than required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:27Z
Check the source to verify that this is the current version before use.

4.4 Tests.-

4.4.1 Stability test.- The type I reel shall be fully loaded with a coil. The loaded strapping coil reel shall by raised on one side of its base to a point of approximate equilibrium and allowed to drop onto the floor. This procedure shall be followed on 4 sides for a rectangular base. The strapping coil reel shall then be examined to determine compliance with 3.5.1.

4.4.2 Maneuverability test.- The type II strapping coil hand truck shall be fully loaded with strapping coil. The loaded strapping coil hand truck shall be pushed over a level floor so that each wheel shall move a minimum of 5 times over a floor obstacle having approximate dimensions of 1/2-inch high by 1-inch wide during a total traveled distance of approximately 100 feet. The strapping coil hand truck during and at the completion of this test shall be examined to determine compliance with 3.5.2.

* 4.4.3 Dispensing test.- A strapping coil reel or hand truck shall be fully loaded with coil, as applicable. Strapping of approximately 10 feet in length shall be drawn and cut-off from the coil. This test shall be repeated 5 times to determine compliance with 3.5.3.

5. PREPARATION

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.-

5.1.1.1 Disassembly.- Each strapping coil reel or hand truck shall be completely disassembled, when feasible. Attaching hardware shall be reassembled onto the frame or placed in a cloth bag or paper envelope.

5.1.1.2 Packaging.- Each strapping coil reel or hand truck shall be packaged in a fiberboard box conforming to style optional, type CF, variety SW, or type SF, class domestic, grade 275 of PPP-B-636. The instruction sheet shall be placed in a paper envelope and packaged in the same box as the strapping coil reel or hand truck. Disassembled components shall be immobilized by the use of fiberboard inserts. Each box shall be securely closed in accordance with the appendix of the container specification.

5.1.2 Level C.- Strapping coil reels or hand trucks shall be packaged to afford adequate protection against deterioration, corrosion, and physical damage during known, favorable conditions, shipment, handling, and storage from the supply source to the first receiving activity. The supplier may use his standard practice, when it meets this requirement.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:27Z
Check the source to verify that this is the current version before use.

5.2.1 Level A.- Each strapping coil reel or hand truck packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style FOL, V2s of PPP-B-636. Alternatively, shipping containers may be used that conform to overseas type of PPP-B-601; class II of PPP-B-591; class 2 of PPP-B-576; class 2, style 2 or 4 of PPP-B-621; or style 1 or 2, class 3, type 2 load of PPP-B-585. Each fiberboard shipping container shall be waterproofed by means of tape in accordance with the appendix of PPP-B-636, and each alternative shipping container shall be provided with a case liner conforming to MIL-L-10547. Closure and reinforcing with flat strapping or tape banding shall be in accordance with the appendix of the applicable container specification.

5.2.2 Level B.- Unless otherwise specified (see 5.2.2.1), each strapping coil reel or hand truck packaged as specified in 5.1.1.2 shall require no further packing except that box closure shall be in accordance with method II of the appendix of the box specification.

5.2.2.1 When specified (see 6.2), the shipping container shall be grade V3c or V3s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the container specification. The shipping container material may also be grade V4s of PPP-F-320.

5.2.3 Level C.- Stands, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Palletization.- Unless otherwise specified (see 6.2), strapping coil reels or hand trucks packed as specified shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Container patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size that does not conform to any of the container patterns specified in MIL-STD-147 the container pattern used shall be approved by the contracting officer.

5.4 Marking.- In addition to any special marking required by the contract or order, interior packages, shipping containers and unitized loads shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The strapping coil reel and hand truck covered by this specification are intended for dispensing coils of flat steel, round steel, flattened wire or nonmetallic strapping used for packaging and packing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:27Z
Check the source to verify that this is the current version before use.

* 6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class reel or hand truck required (see 1.2).
(c) First article.- When preproduction sample is required (see 3.1).
(d) Wheel tires if other than specified (see 3.3.2).
(e) Instruction sheet, contents and distribution (see 3.8).
(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(g) When grade V3c or V3s fiberboard shipping container is required for level B pack (see 5.2.2.1).
(h) When palletization is not required (see 5.3).

6.3 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Air Force - 84

Review activities:

Army - SM
Air Force - 84, 85

Preparing activity:

Army - GL

Project No. 3540-0064

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:27Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Reels, Strapping Coil and Trucks, Hand Strapping Coil MIL-R-43448A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:27Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
U. S. Army Natick Laboratories
Natick, Massachusetts 01760

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Commanding General
U. S. Army Natick Laboratories
ATTN: AMXRES-EQS
Natick, Massachusetts 01760

FOLD

☆ U. S. GOVERNMENT PRINTING OFFICE: 1968-341-514/A-3554

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:27Z
Check the source to verify that this is the current version before use.